

AVINASH KUMAR K S
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Quality Assurance Technician with over **3.5 years of experience in test automation, functional testing, and DevOps practices**, currently working at **Amazon Development Center**. Proficient in **Python, Pytest, Selenium, Appium, and SQL**, with hands-on experience in building scalable automation frameworks, debugging complex test failures, and integrating CI/CD pipelines using **Jenkins and Git**. Demonstrated expertise in **JIRA, TestRail, Confluence**, with a proven track record of leading test initiatives, reducing manual efforts, and enhancing product quality. Recognized for innovation and leadership with multiple accolades at Amazon, including **Quarterback Champion** and **Ace of Innovation**.

TECHNICAL SKILLS

- **Programming languages:** Python, SQL, JAVA (basics)
- **Database:** MySQL
- **Software Testing:** Pytest, Python-Selenium, Appium, Jira
- **Testing Methods & Environment:** Confluence, TestRail, KIBANA, GIT, STLC, Functional, Regression, Ad-Hoc.

CERTIFICATES

- Software Engineer Certificate (HackerRank)

ACHIEVEMENTS

- **Quarterback Champion, Q3 2023** – Recognized for exceptional performance and leadership in driving key initiatives at Amazon.
- **Top Ideator, Ideathon Q4 2023** – Awarded for submitting 24 valid and innovative ideas that contributed to enhancing team processes and innovation.
- **Ace of Innovation, Amazon** – Honored for excellence in automation, continuous improvement, and leading impactful initiatives that advanced operational efficiency.

WORK EXPERIENCE

Amazon Development Center, Chennai
Quality Assurance Technician

May 2022 - January 2026

- Designed and maintained automation test cases, managing codebases with version control, ensuring 100% compliance.
- Debugged 100+ failed automation cases, improving system functionality by 40% and reducing failure rates by 30%.
- Automated 200+ test cases, expanding test coverage by 80%, reducing manual efforts by 70%, and improving execution efficiency by 60%.
- Prepared test documentation, including test plans, reports, and SOPs, strengthening test strategies.
- Triageed defects and analyzed logs to identify root causes, resolving 90% of issues within SLAs and escalating 30+ blocker/critical issues, reducing defect backlog by 25%.
- Conducted integration testing deliverables, ensuring 95% defect detection and reducing post-release defects by 30%.
- Collaborated with cross-functional teams, ensuring smooth User Acceptance Testing (UAT) and validating test verification matrices.
- Led a 10-member team, owning the component end-to-end, set up triage, enabled automation for 100+ apps in batches, and gated critical crashes early.
- Built a script to fetch third-party apps and upload to S3, reducing manual effort by 80% and boosting pipeline test readiness.
- Proposed an AI-driven test report analyzer concept for crash/failure root-causing — adopted and implemented by the SDET team.
- Leveraged Amazon Q and LLM-based tools to generate test cases, debug automation scripts, and optimize test workflows.
- Designed workflows for AI-driven test case generation from UI inputs, improving test coverage and reducing manual effort.

Amazon Development Center, Coimbatore
Quality Associate

August 2020 - May 2022

- Handled a high volume of customer chats and calls daily, ensuring quick and accurate support even during peak times.
- Focused on resolving issues on the first interaction, improving customer satisfaction and repeat contacts.

- Recognized with an RNR award for consistent performance, quality, and commitment to customer experience.

PROJECTS

failureIQ (Featured)

github.com/akaAvinash/failureIQ

- AI-powered QA investigation engine that automatically root-causes failing tests using the Claude API, layered on top of any test automation framework.
- Framework-agnostic plugin architecture supporting pytest/Playwright, Robot Framework, and generic JSON ingest. MIT licensed.

Tech Stack: Python, Claude API, Pytest, Playwright

DeviceOps (In Progress)

github.com/akaAvinash/DeviceOps

- Simulated OTA/device-fleet management system built end-to-end with a FastAPI + SQL backend.
- Test coverage via Playwright, pytest, API tests, and k6 load testing, with OCR validation; AI self-healing/failure-summary layer planned.

Tech Stack: Python, FastAPI, Playwright, k6, Claude API

Interview Agent

github.com/akaAvinash/interview-agent

- Autonomous interview-prep agent that generates tailored mock interviews from a resume and job description.
- Conducts interviews conversationally and scores answers via LLM-as-judge.

Tech Stack: Python, FastAPI, Streamlit, Claude API

homelab-agent

github.com/akaAvinash/homelab-agent

- Self-hosted AI monitoring and auto-remediation agent combining a local LLM with Claude API escalation.
- Safety-first, whitelist-only execution model — AI never runs arbitrary commands directly.

Tech Stack: Python, Local LLM, Claude API, Docker

tplink-vpn-automation

github.com/akaAvinash/tplink-vpn-automation

- Playwright-based browser automation for a router VPN setting, deployed as an unattended scheduled service.
- Layered retry logic and file-locking for reliability.

Tech Stack: Python, Playwright, Automation

ADDITIONAL SKILLS

- Experience in designing and executing test cases, tracking nonconformance items, and ensuring compliance before production rollout.
- Strong analytical skills for process monitoring, product evaluation, and reporting quality metrics.
- Effective communication and problem-solving abilities, ensuring smooth collaboration across teams.

EDUCATION

Kerala Technical University, India

Bachelors of Technology in Mechanical Engineering

August 2015 - May 2019

CGPA 6.64